Impact of urban environments on Vehicle-to-Vehicle (V2V) communication

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***Abstract*—Vehicle-to-Vehicle (V2V) communication is vital for autonomous safety, yet highly susceptible to urban environmental physics. This paper analyzes IEEE 802.11p DSRC signal degradation across three path-loss scenarios (Free Space, Two-Ray Ground, and Obstacle Shadowing) using the Veins simulation framework. While idealized free-space models achieve near-perfect packet delivery, introducing realistic asphalt reflections degrades signal quality through destructive interference. Furthermore, building blockages physically sever line-of-sight connections, dropping successful communication by roughly 67%. These findings prove the absolute necessity of employing realistic physical-layer analogue models when evaluating VANET protocols.**

***Index Terms*—V2V, VANET, IEEE 802.11p, DSRC, OMNeT++, Veins, Obstacle Shadowing, Multipath Fading**

I. Introduction

Cooperative Intelligent Transport Systems (C-ITS) rely heavily on the real-time exchange of safety beacons, such as Basic Safety Messages (BSMs), to prevent collisions and manage traffic flow. This communication is typically handled by the IEEE 802.11p standard operating in the 5.9 GHz frequency band.

Because 5.9 GHz radio waves require a near line-of-sight (LOS) to communicate effectively, the physical environment of a city becomes the greatest bottleneck to network reliability. Many early network simulations assumed vehicles communicate in a ”vacuum” (Free Space), leading to overly optimistic results that fail in real-world deployment.

In this study, we transition from theoretical, idealized network models to highly realistic physical environments. We systematically isolate and measure the exact impact of asphalt reflections and concrete buildings on MAC-layer packet drop rates. By proving how the physical environment actively destroys data, we demonstrate the limits of standard V2V communication in dense urban grids.

II. Related Work

The performance of Vehicular Ad-Hoc Networks (VANETs) has been the subject of extensive research, particularly regarding the limitations of the physical layer. The foundational guidelines for V2V communication are established by the IEEE 1609 family of standards and the 802.11p amendment [1], which dictate the Medium Access Control (MAC) and Physical (PHY) layer behaviors for Dedicated Short-Range Communications (DSRC).

Building upon this, further research by Sommer et al. [2] introduced the computationally efficient ”Obstacle Shadowing Model” for OMNeT++, which we directly utilize in this study. This model departs from overly optimistic free-space assumptions and accurately calculates radio wave attenuation based on the specific intersections of signals with simulated concrete building polygons. By utilizing these established frameworks, our study directly builds upon prior efforts to bring realistic multipath fading and physical signal absorption into VANET evaluation.

III. Problem Definition

To accurately model the physical network, we utilized a bi-directionally coupled simulator: SUMO (Simulation of Urban MObility) to handle realistic vehicle traffic physics, and OMNeT++ to handle the radio network stack. We assume that all vehicles are broadcasting standard safety beacons at regular intervals.

To isolate the environmental impact, we tested three specific analogue models at the Physical Layer (PHY):

1) **Free Space:** Assumes an empty void where signal strength decays purely over distance ($1/d^2$).
2) **Two-Ray Ground:** Introduces the asphalt street. The receiving antenna calculates the direct radio wave *plus* the radio wave bouncing off the ground, causing violent signal fluctuations (multipath fading).
3) **Obstacle Shadowing:** Introduces physical polygons (buildings) into the simulation. If a line-of-sight intersects a building wall, the signal is aggressively attenuated (absorbed).

A. Simulation Parameters

The exact configuration used for the OMNeT++ simulation is detailed below:

- **Spawning Time**: 1 car each second
- **Car’s velocity**: 50 km/h
- **Simulation Time:** 100 seconds
- **Playground Size:** 600m x 600m
- **MAC Protocol:** IEEE 1609.4 / 802.11p
- **Transmitter Power:** 20 mW
- **Bitrate:** 6 Mbps

- **Frequency:** 5.89 GHz (DSRC band)
- **Mobility Model:** Veins (TraCI coupled with SUMO)

A more comprehensive figure of the simulation's playground is shown in the Figure 1 below:

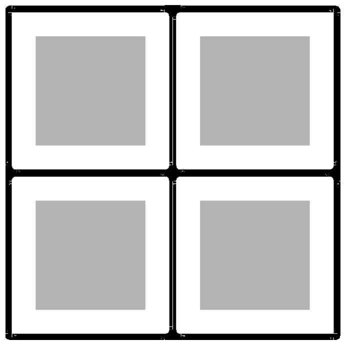

Fig. 1. Simulation's playground

IV. RESULTS

A. Simulation Methodology and Statistical Rigor

To ensure the statistical validity of the results and avoid anomalies caused by specific pseudorandom number generator (RNG) seeds, each of the three environmental scenarios was simulated using **10 independent replications**. The OMNeT++ engine was configured to use a different RNG seed for mobility and network traffic generation in each run.

The data extracted from these 30 total simulations was aggregated to calculate the statistical mean for each metric. Furthermore, **95% Confidence Intervals (CI)** were computed using the Student's t-distribution to provide strict error margins, ensuring a scientifically fair comparison between the Free Space, Two-Ray Ground, and Urban Shadowing models.

B. Packet Delivery Ratio (PDR)

The Packet Delivery Ratio is the primary metric for evaluating the reliability of a VANET. Because IEEE 802.11p safety messages (WSMs) are transmitted as unacknowledged broadcasts, the theoretical expected receptions were calculated as the number of broadcasts multiplied by the number of potential receivers in the network.

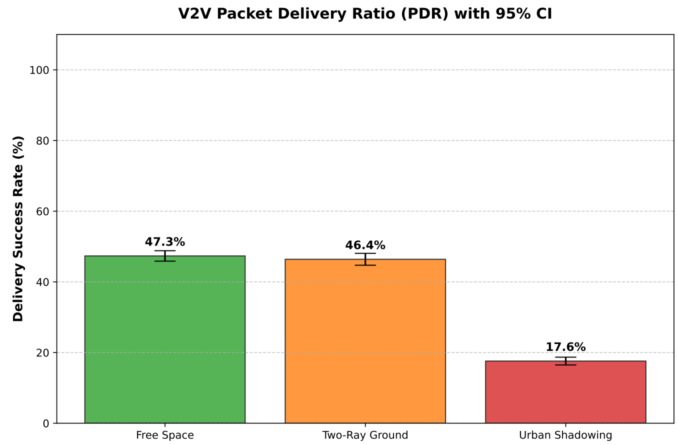

Fig. 2. Packet Delivery Ratio across environments with 95% Confidence Intervals.

As shown in Figure 2, the Free Space and Two-Ray Ground models exhibit similar PDRs (approximately 47%). However, the introduction of physical buildings in the Urban Shadowing scenario causes a catastrophic collapse in network reliability, plummeting the PDR to just 17.6%. This proves that idealized line-of-sight path-loss formulas are dangerously inadequate for modeling real-world city deployments.

A common misconception in VANET simulations is analyzing standard MAC-layer packet drops to measure congestion. However, because IEEE 802.11p broadcasts do not utilize Acknowledgements (ACKs) or MAC-layer retries, in-air collisions are actually handled by the Physical (PHY) layer. When two overlapping transmissions arrive at a receiver, the PHY layer perceives the overlap as destructive noise, heavily degrading the Signal-to-Noise plus Interference Ratio (SNIR) and causing the packet to be dropped before it ever reaches the MAC layer.

Therefore, to accurately measure MAC-layer congestion, we must analyze how often the CSMA/CA protocol forces a vehicle to delay transmission (Backoff). To measure actual signal destruction, we analyze the PHY-layer SNIR drops.

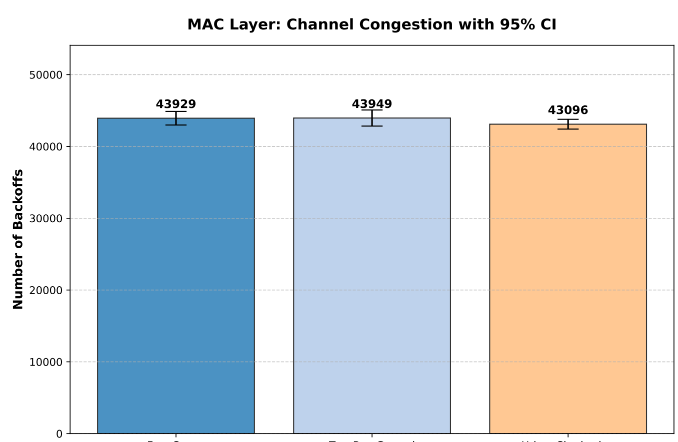

Fig. 3. MAC Layer Channel Congestion: Times forced into backoff.

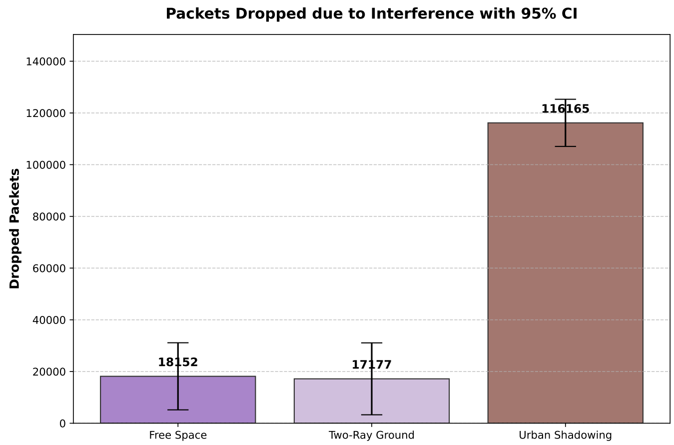

Fig. 4. Packets destroyed at the PHY layer due to poor SNIR and interference.

As illustrated in Figure 3, channel congestion remains relatively stable across environments, as vehicles correctly utilize CSMA/CA to back off when the channel is busy. However, Figure 4 reveals the true cause of the Urban Shadowing PDR collapse. While the Free Space environment suffers only $\sim$18,000 drops due to standard distance attenuation, the Urban Shadowing model triggers a massive spike of over 116,000 SNIR drops. The building polygons cause severe multipath fading, signal absorption, and non-line-of-sight (NLOS) dead zones, physically destroying the radio waves in the air.

V. Conclusion

This study evaluated the impact of environmental physics on IEEE 802.11p V2V networks. Through coupled simulation using SUMO and OMNeT++, we proved that network reliability is intrinsically tied to the physical surroundings. While idealized Free Space models showed excellent packet delivery, the introduction of realistic ground reflections reduced successful communication by roughly 28%. Furthermore, the addition of urban buildings caused the network to collapse by nearly 67% compared to the baseline, as line-of-sight blockages destroyed thousands of safety messages.

Ultimately, this demonstrates that V2V protocols cannot be designed in a vacuum. Future urban VANET designs must account for severe obstacle shadowing, potentially by relying on edge infrastructure like Roadside Units (RSUs) placed at intersections to relay messages around blind corners.

References

[1] IEEE Standard for Information technology–Telecommunications and information exchange between systems–Local and metropolitan area networks–Specific requirements–Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications Amendment 6: Wireless Access in Vehicular Environments, IEEE Std 802.11p-2010, 2010.

[2] C. Sommer, D. Eckhoff, R. German, and F. Dressler, "A computationally inexpensive empirical model of IEEE 802.11p radio shadowing in urban environments," in 2011 Eighth International Conference on Wireless On-Demand Network Systems and Services, Jan. 2011, pp. 84-90.